Merge Sort

Topic: DivideAndConquer File: d1.pdf

PROBLEM

Sort an array in nondecreasing order using divide and conquer.

DIVIDE

Split the array into two halves until each subarray contains zero or one element.

CONQUER AND COMBINE

Recursively sort both halves, then merge the two sorted halves by repeatedly taking the smaller front element.

PSEUDOCODE

```
mergeSort(A):
  if length(A) <= 1: return A
  split A into L and R
  return merge(mergeSort(L), mergeSort(R))
```

EXAMPLE

[7, 2, 6, 3] splits into [7,2] and [6,3], then merges to [2,3,6,7].

COMPLEXITY

Time: O(n log n) in all cases. Auxiliary array space: O(n).